

COURSE NUMBER: CM1400

COURSE TITLE: Technical Report Writing I

COURSE DESCRIPTION:

This course is designed to teach technology students the fundamentals of technical reporting. Emphasis is on strategies of technical reporting, research techniques and organizational skills.

PREREQUISITES: None

CO-REQUISITES: None

CARNAGIE UNITS: Three (3)

CREDIT VALUE: Three (3)

COURSE HOURS PER WEEK: Three (3)

LAB HOURS PER WEEK: Zero (0)

REQUIRED TEXT:

Lannon, J. M., Gurak, L. J., Klepp, D., Kelly, S. (2021). *Technical communication*. (9th Canadian edition). Pearson. eText ISBN: 9780137967742

LEARNING RESOURCES: None

MAJOR TOPICS:

- 1.0 Sentence and Paragraph Construction
- 2.0 Characteristics of Technical Writing
- 3.0 Technical Descriptions
- 4.0 Research Techniques
- 5.0 Technical Abstract
- 6.0 Writing in a Digital Environment

LEARNING OBJECTIVES:

1.0 Sentence and Paragraph Construction

- 1.1 Examine types of sentences including:
 - 1.1.1 Interrogative

- 1.1.2 Declarative
- 1.1.3 Imperative
- 1.1.4 Exclamatory
- 1.1.5 Simple
- 1.1.6 Compound
- 1.1.7 Complex
- 1.1.8 Compound-Complex
- 1.2 Identify the essential parts of a sentence including:
 - 1.2.1 Subject
 - 1.2.2 Verb/Predicate
 - 1.2.3 Direct object
 - 1.2.4 Indirect Object
- 1.3 Differentiate among phrases, clauses, and sentences
- 1.4 Differentiate between dependent and independent clauses
- 1.5 Analyze common sentence errors including:
 - 1.5.1 Fragments
 - 1.5.2 Comma splices
 - 1.5.3 Fused sentences
- 1.6 Apply strategies for correcting common sentence errors
- 1.7 Analyze common sentence errors which contribute to unclear messages including:
 - 1.7.1 Unnecessary shifts
 - 1.7.2 Pronoun references
 - 1.7.3 Parallel structures
- 1.8 Analyze major concepts related to subject-verb agreement
- 1.9 Write clear, concise, complete sentences which adhere to the conventions of grammar, punctuation, and mechanics
 - 1.9.1 Use punctuation marks including:
 - 1.9.1.1 Colons
 - 1.9.1.2 Semicolons
 - 1.9.1.3 Commas
- 1.10 Apply principles of paragraph construction
- 1.11 Discuss the basic purposes for writing
- 1.12 Describe the major characteristics of an effective paragraph
- 1.13 Analyze the following in selected paragraphs:
 - 1.13.1 Intended audience
 - 1.13.2 Primary purpose
 - 1.13.3 Topic sentence
- 1.14 Apply the following steps in the writing process:
 - 1.14.1 Planning
 - 1.14.2 Shaping

- 1.14.3 Drafting
- 1.14.4 Revising
- 1.14.5 Editing
- 1.14.6 Proofreading
- 1.15 Apply techniques for gathering ideas including:
 - 1.15.1 Free writing
 - 1.15.2 Brainstorming
 - 1.15.3 Outlining
 - 1.15.4 Mapping
- 1.16 Examine selected paragraphs for key elements including:
 - 1.16.1 Details
 - 1.16.2 Coherence techniques
 - 1.16.3 Sentence arrangement
 - 1.16.4 Methods of development
- 1.17 Write well-developed paragraphs which illustrate the following:
 - 1.17.1 A variety of sentence arrangements
 - 1.17.2 Conciseness
 - 1.17.3 Clarity
 - 1.17.4 Coherence
 - 1.17.5 Unity
 - 1.17.6 Correct sentence structure
 - 1.17.7 Correct grammar
 - 1.17.8 Correct punctuation
 - 1.17.9 Appropriate use of mechanics

2.0 Characteristics of Technical Writing

- 2.1 Evaluate the elements of effective technical writing including:
 - 2.1.1 Reader-centred focus
 - 2.1.2 Purpose
 - 2.1.3 Audience type
 - 2.1.4 Levels of technicality
 - 2.1.5 Positive tone
 - 2.1.6 Technical style
- 2.2 Distinguish between writing climactically and writing technically
 - 2.2.1 Discuss the methods for reading technical documents
 - 2.2.2 Justify how note-taking skills help focus on main ideas
 - 2.2.3 Differentiate between informal and formal sentence definitions
 - 2.2.4 Write formal sentence definitions from a list of related terms
 - 2.2.5 Explain the methods by which definitions may be incorporated into a report

3.0 Technical Descriptions

- 3.1 Outline the guidelines for describing a mechanism
 - 3.1.1 Write a physical description of a small mechanism from a given list for an uninformed reader of a mechanism at rest or a mechanism in operation
- 3.2 Distinguish between describing a mechanism and describing a mechanism in operation
- 3.3 Write a description of a mechanism in operation from a list of topics
- 3.4 Define process description
 - 3.4.1 Explain how a process description focuses more on what the technician does rather than on how the instruments work
 - 3.4.2 Outline the sequence for a process description
 - 3.4.3 Describe the process for a given topic

4.0 Research Techniques

- 4.1 Discuss methods of data collection including:
 - 4.1.1 Observation
 - 4.1.2 Testing
 - 4.1.3 Interviewing
 - 4.1.4 Studying printed material
- 4.2 Apply techniques for gathering ideas including:
 - 4.2.1 Free writing
 - 4.2.2 Brainstorming
 - 4.2.3 Outlining
 - 4.2.4 Mapping
- 4.3 Apply the steps in the writing process including:
 - 4.3.1 Planning
 - 4.3.2 Shaping
 - 4.3.3 Drafting
 - 4.3.4 Revising
 - 4.3.5 Editing
 - 4.3.6 Proofreading
- 4.4 Select a topic for a mechanism in operation or for a process
 - 4.4.1 Research the topic
 - 4.4.2 Record the findings
 - 4.4.3 Develop an outline for a topic
- 4.5 Assess the value of using audio/visual graphics in a technical presentation
 - 4.5.1 Create a document utilizing audio/visual graphics

5.0 Technical Abstract

- 5.1 Analyze an article from a technical journal
 - 5.1.1 Explain the main ideas from the article
 - 5.1.2 Identify the topic sentence of each paragraph
 - 5.1.3 List the summary statements of the article
 - 5.1.4 Define technical terms in the article
- 5.2 Write a draft of the technical abstract highlighting the major points
 - 5.2.1 Insert appropriate changes to the rough draft including:
 - 5.2.1.1 Appropriate sentence structure
 - 5.2.1.2 Paragraph structure
 - 5.2.1.3 Mechanics
 - 5.2.2 Reduce the final draft of the technical abstract to one quarter to one tenth of the original size

6.0 Writing in a Digital Environment

- 6.1 Analyze on-line media sites (such as blogs/wikis) for technical communication including the audience and purpose
 - 6.1.1 Assess readers' expectations when reading digital content
 - 6.1.2 Evaluate the following when writing for a digital audience including:
 - 6.1.2.1 Ethical
 - 6.1.2.2 Legal
 - 6.1.3 Create a professional on-line presence
- 6.2 Analyze the audience and purpose when writing workplace emails
 - 6.2.1 Discuss the components of a workplace email message
 - 6.2.2 Recommend how to utilize the subject line to persuade an audience to open messages
 - 6.2.3 Write emails in a professional style and tone for a global audience

EVALUATION:

Note: Assignments should be tailored to align with the students' program of study.

Tests:	25%
Assignments:	
Technical Description:	20%
Technical Process Description:	15%
Technical Abstract:	15%
Technical Email:	10%
Professional On-Line Presence:	15%

DATE DEVELOPED: June 1994

DATE REVIEWED: November 2018

REVISION NUMBER: 10

DATE REVISED: October 2025

Effective PeopleSoft Date: September 2026

Note to instructor: Check PIRS to ensure this outline is the most current version.